



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

8.0 Introduction to Digital Twin

9.0 Digital Twin Architecture

10.0 Data Communication in Digital Twin

11.0 Digital Twin Applications

July 2025

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- 11.2 Use cases of Digital Twin in O&G
- 11.3 How Digital Twin Contributes to Sustainability

Learning Outcome

At the end of the session, you will be able to:

- Describe concept of digital twin technology and its architecture
- Explain the role of digital twin in shaping the industrial needs
- Assess the real-life applications to evaluate the opportunities in digital twin

SKG14 TI&R Recap

7-3-13-0003-QR2 v3.0

SKG 14 – BASE Technology Competency Matrix / Descriptor

Discipline – Instrument & Control

Sub-Discipline – Instrument & Control

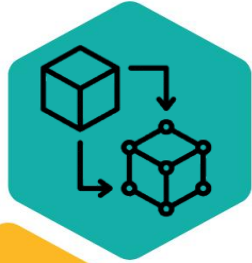
Effective Date – 01 July 2025



PETRONAS

TECHNOLOGY INVENTORY		COMPETENCE LEVELS				
		Awareness	Knowledge	Skill	Advanced	Expert
P5	 Digital Twin PETRONAS definition of a digital twin is the "virtual" and connected representation of equipment, process or system. A digital replica of physical assets at various levels e.g. equipment, package, unit or entire facilities. The digital twin allows all manners of manipulation and optimization to be simulated during design, engineering, construction, and operational phase of assets.  Digitalization  Commercialization <td> Describe digital twin definition, types and its application. Explain the infrastructure and technology to enable digital twin. </td> <td> Specify requirement for digital twin implementation. Describe the integration requirement to enable digital twin. </td> <td> Provide input during multidisciplinary review to enable digital twin implementation. Participate in digital twin implementation related to equipment or asset. </td> <td> Mentoring, coaching or training (within OPU or SKG) </td> <td> Shaping the development and direction of industry and/or academic institution. </td>	Describe digital twin definition, types and its application. Explain the infrastructure and technology to enable digital twin.	Specify requirement for digital twin implementation. Describe the integration requirement to enable digital twin.	Provide input during multidisciplinary review to enable digital twin implementation. Participate in digital twin implementation related to equipment or asset.	Mentoring, coaching or training (within OPU or SKG)	Shaping the development and direction of industry and/or academic institution.

8.0 Introduction to Digital Twin (DTw)



8.1

Evolution of Digital Twin



8.2

Concept of Digital Twin



8.3

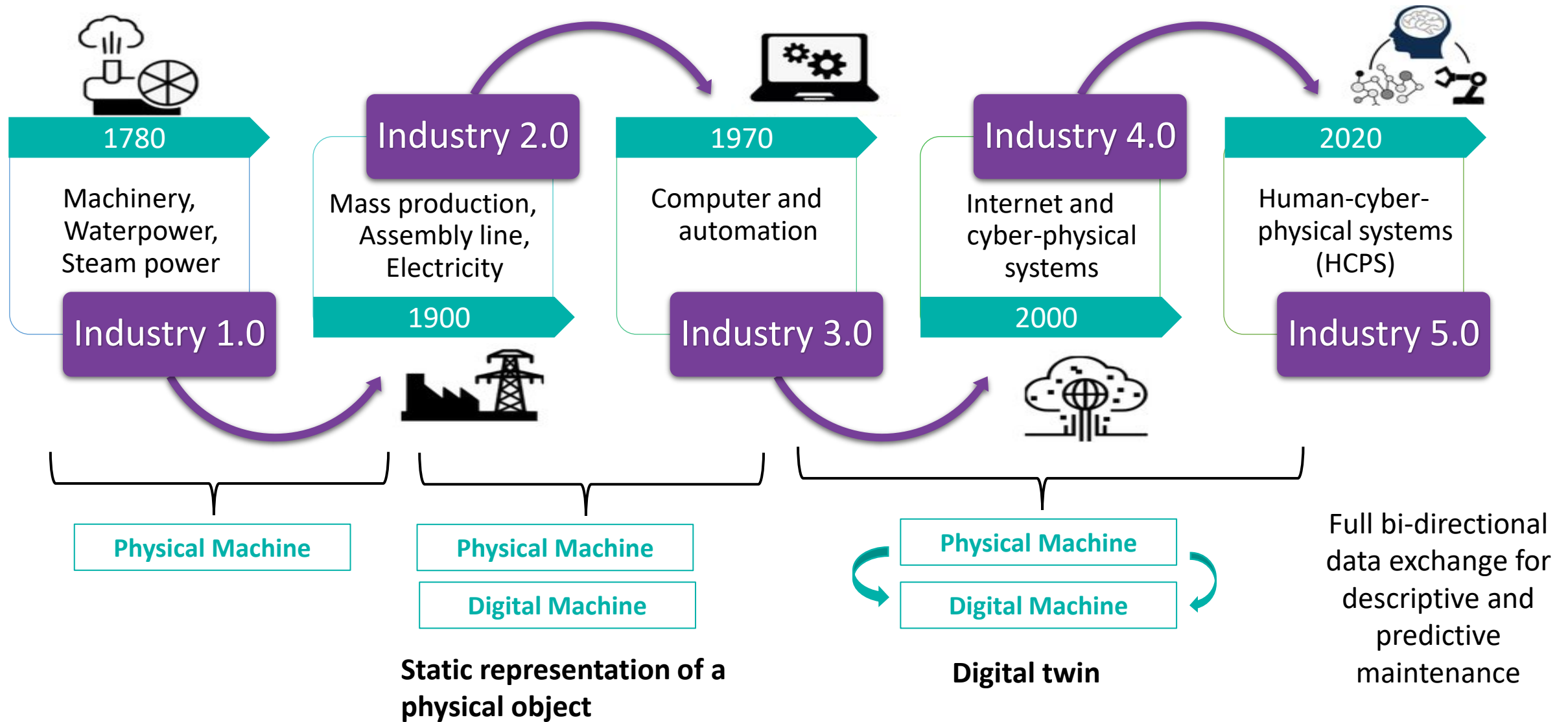
International Standards, i.e.,
ISO 10303, ISO 23247, IEC 30173



8.4

Digital Twins Levels and Types

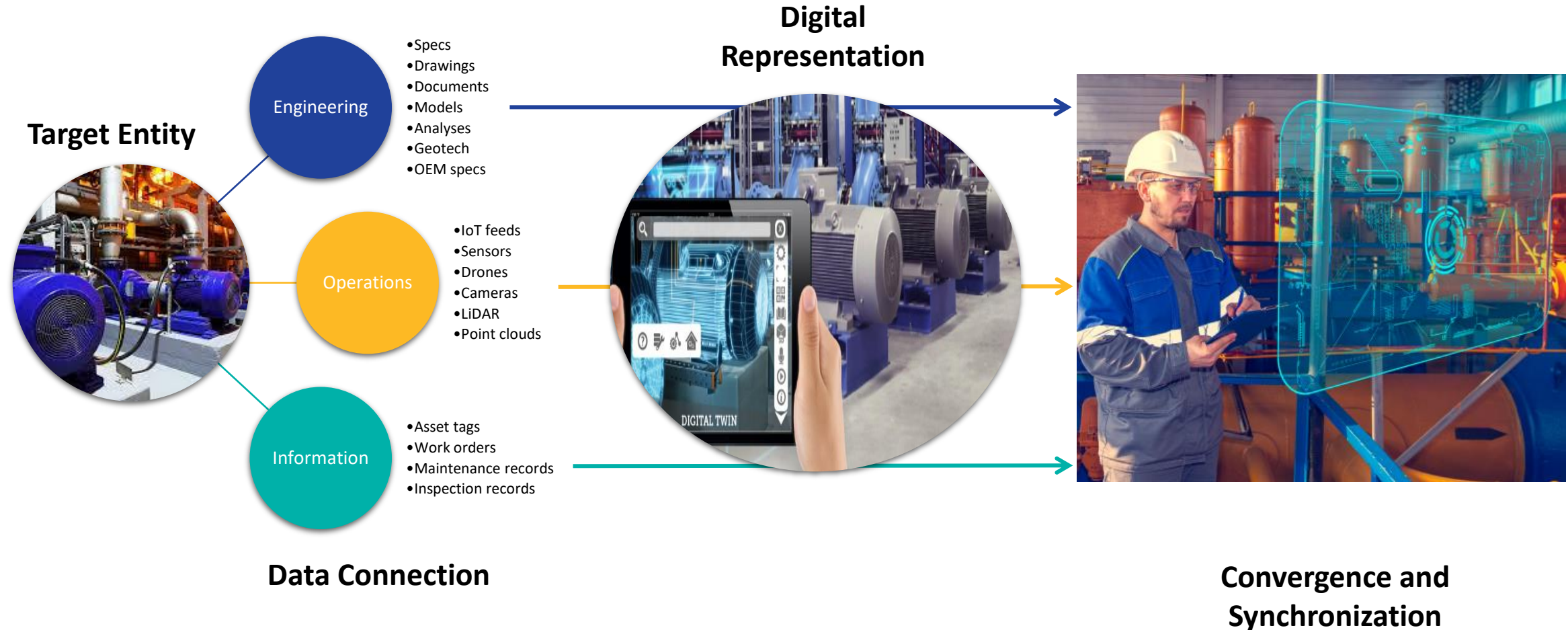
8.1 Evolution of Digital Twin



8.2 Concept of Digital Twin

What is Digital Twin?

Digital representation of a **target entity** with **data connections** that enable **convergence** between the **physical** and **digital** states at an appropriate rate of **synchronization**.



8.2 Concept of Digital Twin

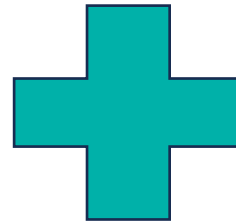
Digital Twin definition in PETRONAS:

"Digital Twins are digital representations of physical processes or systems, with the objective to understand the current and future state, and to allow response in an agile and effective manner to improve business operations. Digital Twins shall be VIRTUAL and CONNECTED representations of equipment, processes or systems."



VIRTUAL

A full or partial "MODEL" of the equipment, process or system is available in a digitized manner.



CONNECTED

The model is connected to real-time (or near real-time) sensors allowing for the model to reflect actual conditions.

8.3 International Standards, i.e., ISO 10303, ISO 23247, IEC 30173

IEC 30173

- terms and definitions of digital twin
- concepts of digital twin,
- functional view of digital twin, and digital twin stakeholders

Standard for
digital twin
terminology

ISO 23247

- Partitioning a digital twin system by 4 layers
- User Entity
- Core Entity
- Data Collection and Device Control Entity
- Observable manufacturing elements (describes the items on the manufacturing floor that need to be modelled)

Standard for digital
twin framework

ISO 10303

ISO 10303-1

- representing product information in a computer-interpretable format
- enables consistent product descriptions across different systems
- file exchange, product databases and long-term data archiving

Standard for Virtual
Entity and Data
representation

8.4 Digital Twin Types

Digital Twin Type according to IEC 30173

Process Digital Twin

- Provides view into a set of activities or operations.
- Consist of a set of physical entities or system digital twins but focuses more on the process itself rather than the physical entities.

System Digital Twin

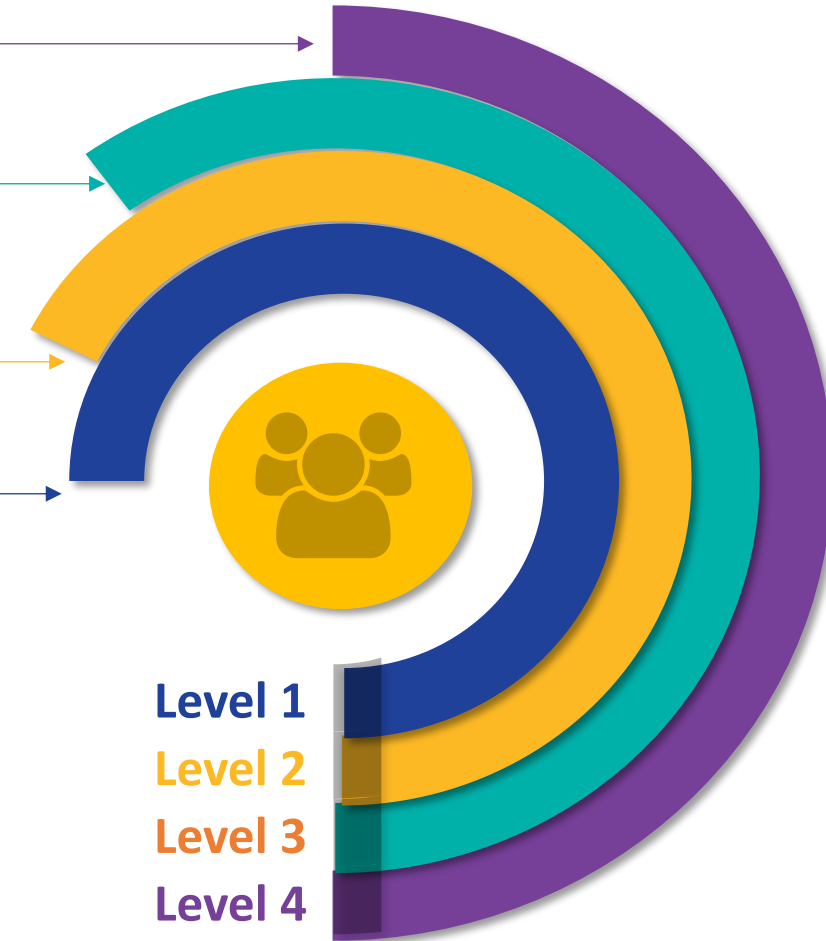
- Collection of target entities and digital entities that together perform a system- or network-wide function.
- Provides visibility into set of interconnected or interdependent target entities.

Asset Digital Twin

- Collection of component digital twins and consist of component digital twin
- Provide visibility at the unit level.

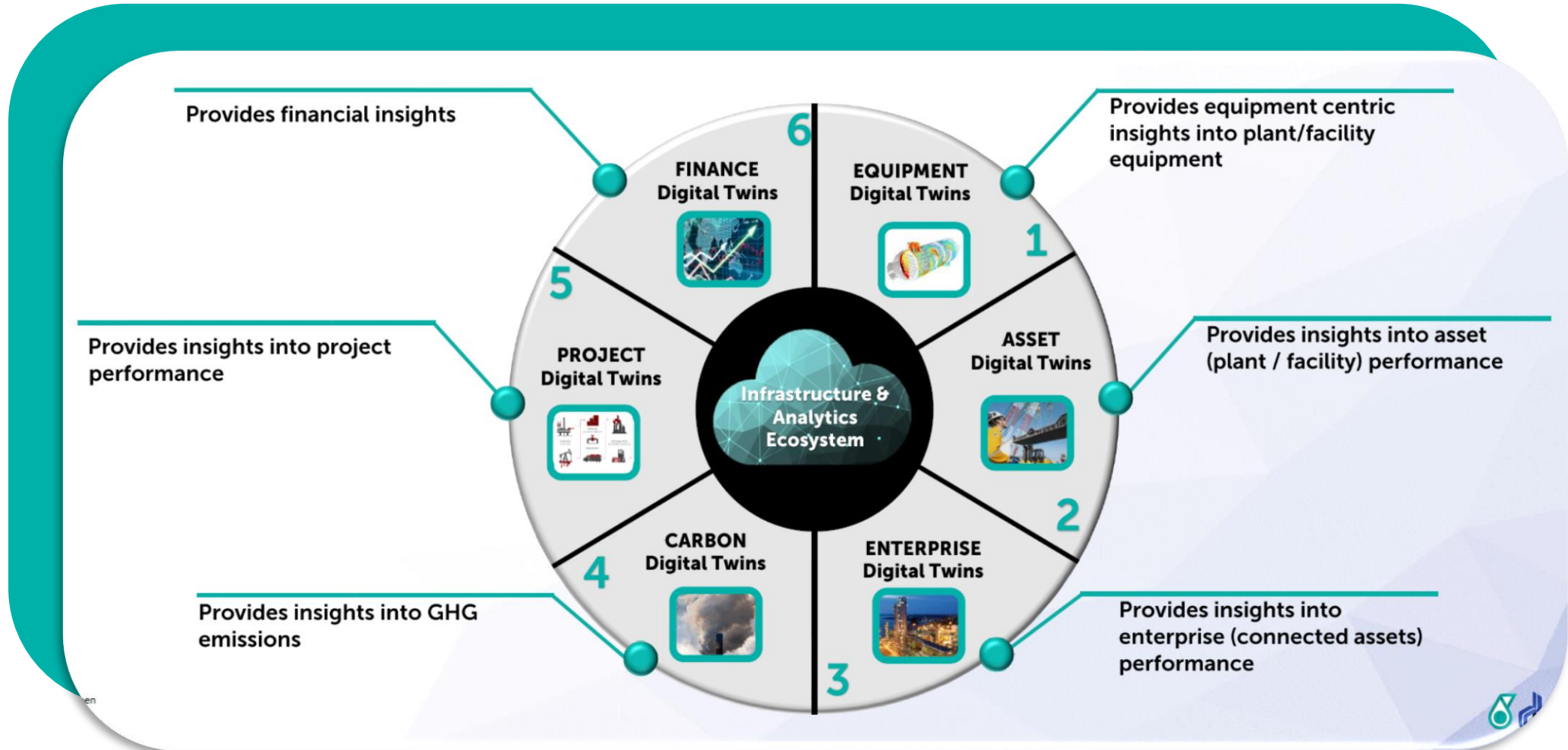
Component Digital Twin

- Major element that has a significant impact on the performance of the target entity to which it belongs.
- Related to the level of complexity within the system.



8.4 Digital Twin Types

Digital twin Type and roles according to PETRONAS



8.4 Digital Twin Maturity Level

Digital Twin Level 1 Descriptive Digital Twins

- Provides real time/near-real time representation of an equipment, process or system

Digital Twin Level 2 Predictive Digital Twins

- Provides descriptive abilities and prediction of future conditions and/or events

Digital Twin Level 3 Prescriptive Digital Twins

- Provides descriptive/predictive abilities and provides recommendations to promote positive situations while avoiding others

9.0 Digital Twin Architecture



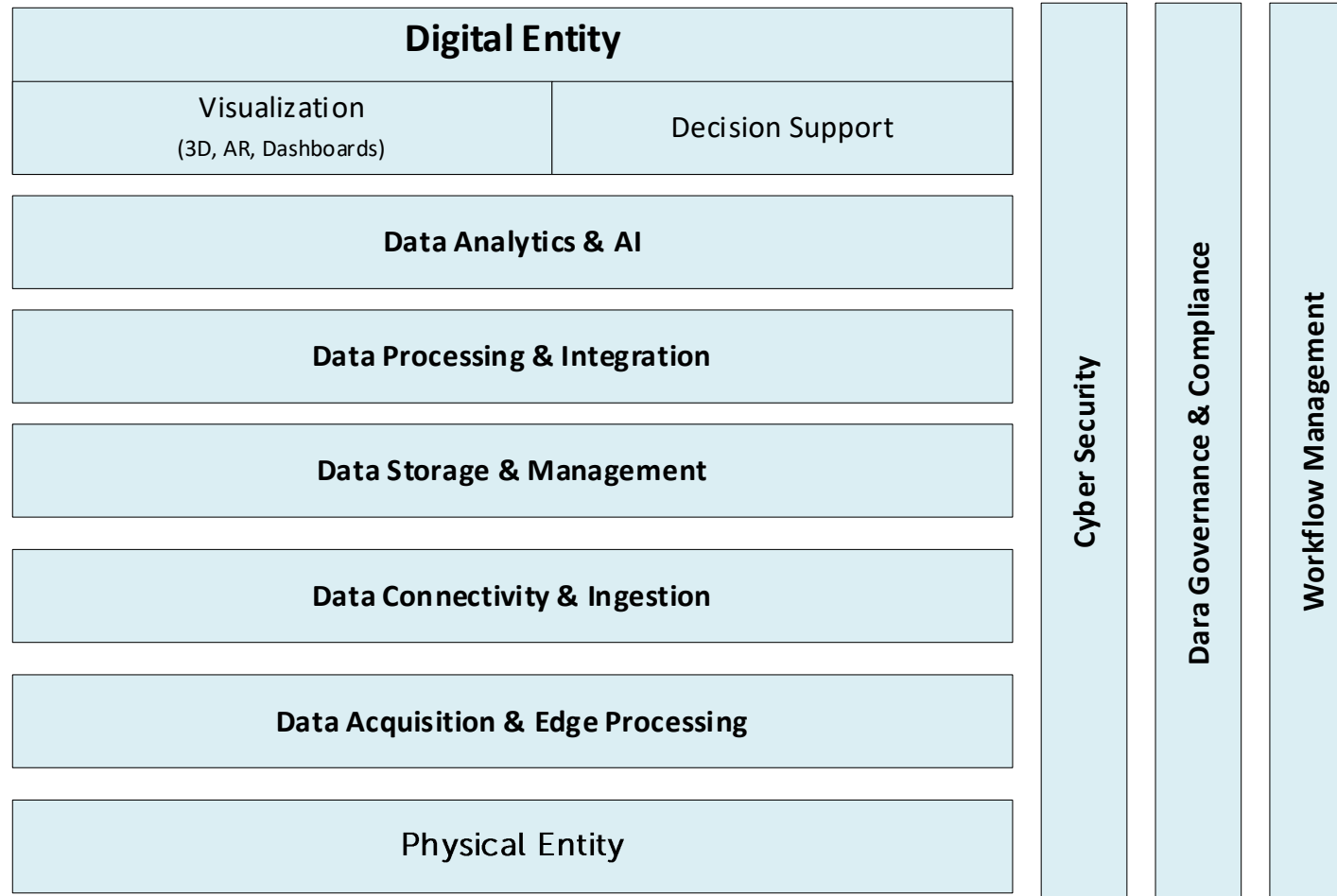
9.1 Model Architecture: ISO 23247-2



9.2 Architecture Core Components

9.0 Digital Twin Architecture

9.1 Model Architecture



Source: GTS

9.0 Digital Twin Architecture

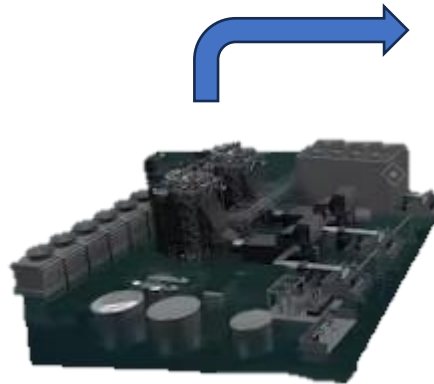
9.2 Core Component

Core Components:

1. Physical entity(real-world system).
2. Digital entity (virtual model).
3. Data connection layer (IoT sensors, cloud).
4. Data analytics & AI for predictive capabilities.
5. Integration with simulation tools and data visualization platforms.

3. Data Connection Layer

- Objects
- Processes
- System



1. Physical entity

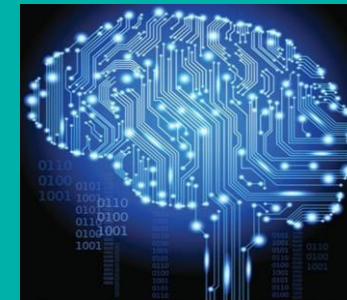
- Objects
- Processes
- System



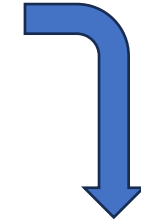
4. Data Analytics and AI



Data collection



5. Integration - Information and decision



2. Digital entity

- Representation models
- Data management
- Data analytics



9.0 Digital Twin Architecture

9.2.1 Physical Entity

What is Physical Asset/Target Entity?

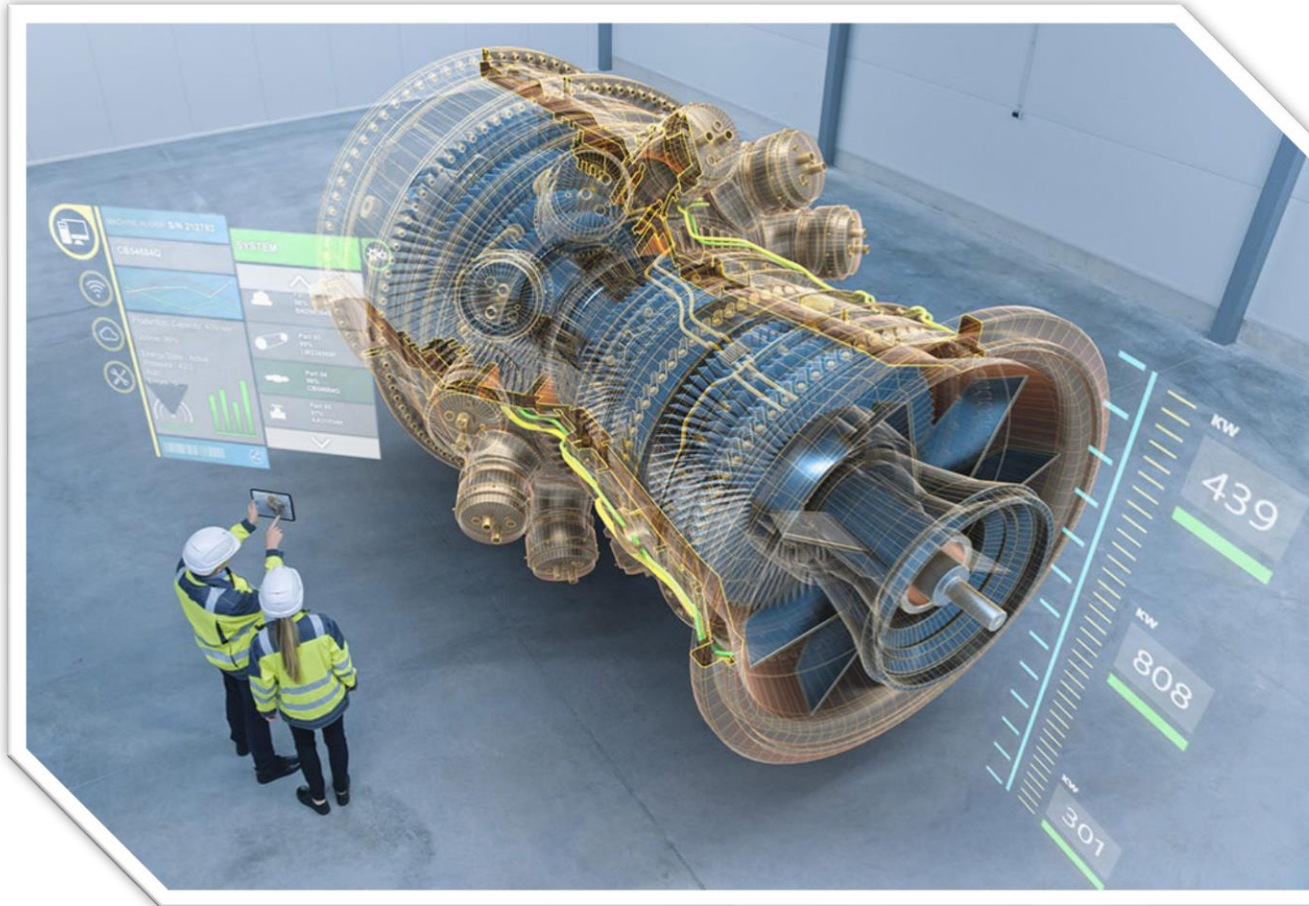
The Physical Asset/Target Entity refers to the **real-world object** or system being represented, monitored, and analyzed by its virtual counterpart.

This can include **machinery, infrastructure, devices, or processes** within industries like energy, manufacturing, or transportation.



9.0 Digital Twin Architecture

Digital/Virtual Entity

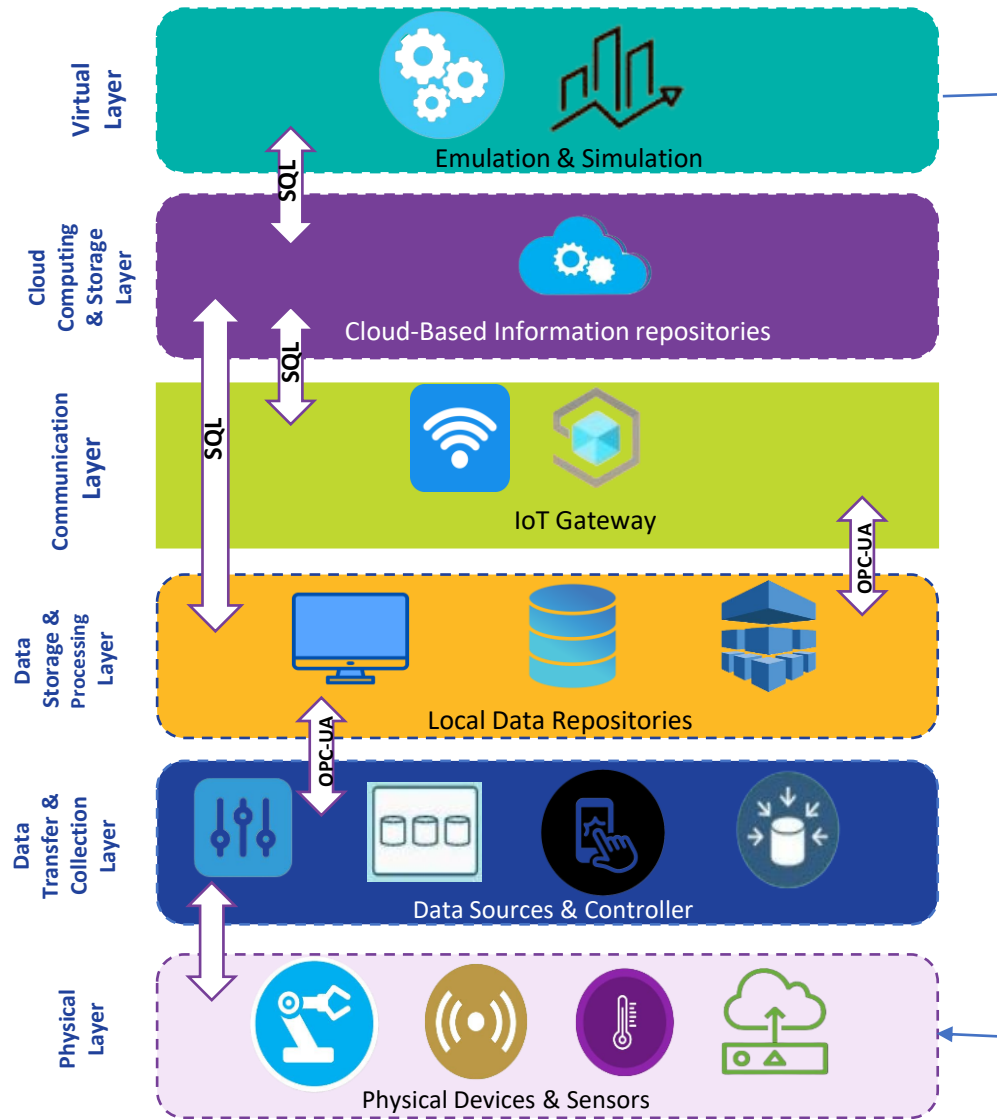


Definition:

A Digital Entity represents the **virtual counterpart** of a **physical asset, process, or system**. It mirrors real-time data and behavior, enabling monitoring, simulation, and optimization.

- Real-time synchronization with the physical counterpart
- Predictive analytics and simulation capabilities
- Enhanced decision-making through scenario testing

Data connection layer (IoT sensors, clouds)



The device communication domain monitors and collects data from sensor devices in the observable manufacturing domain and controls and actuates devices in the observable manufacturing domain. The device communication domain links OMEs to their digital twins for synchronization

Data Analytics and AI – Enabler for descriptive and prescriptive

Descriptive Analytics

Descriptive analytics focuses on summarizing historical data to understand the past.

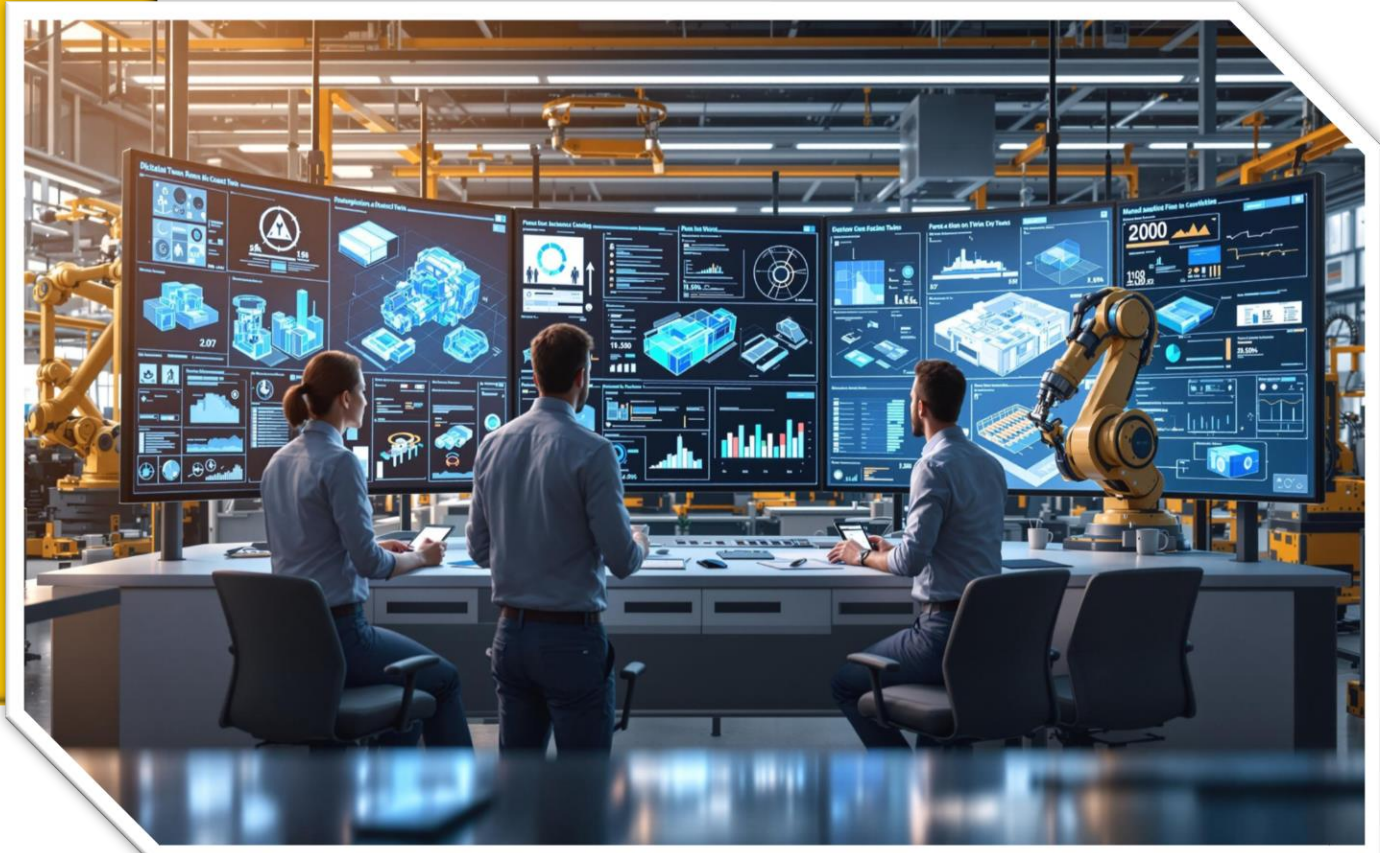
Purpose: It helps identifying trends, patterns, and insights from past events

Prescriptive Analytics

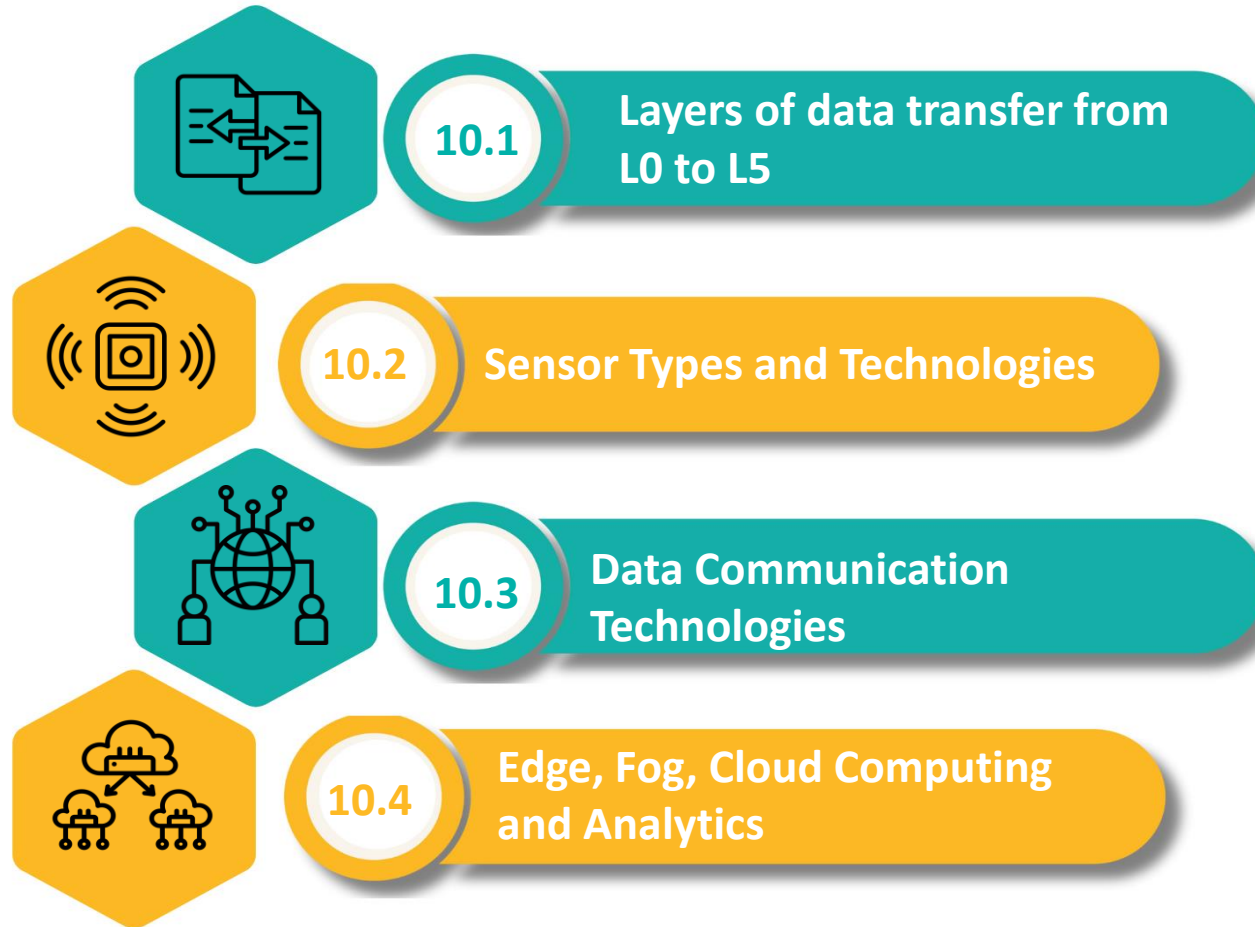
Prescriptive analytics goes beyond predicting future outcomes and provides actionable recommendation for decision-making

Purpose: It suggests the best course of action to achieve a desired outcome based on data analysis

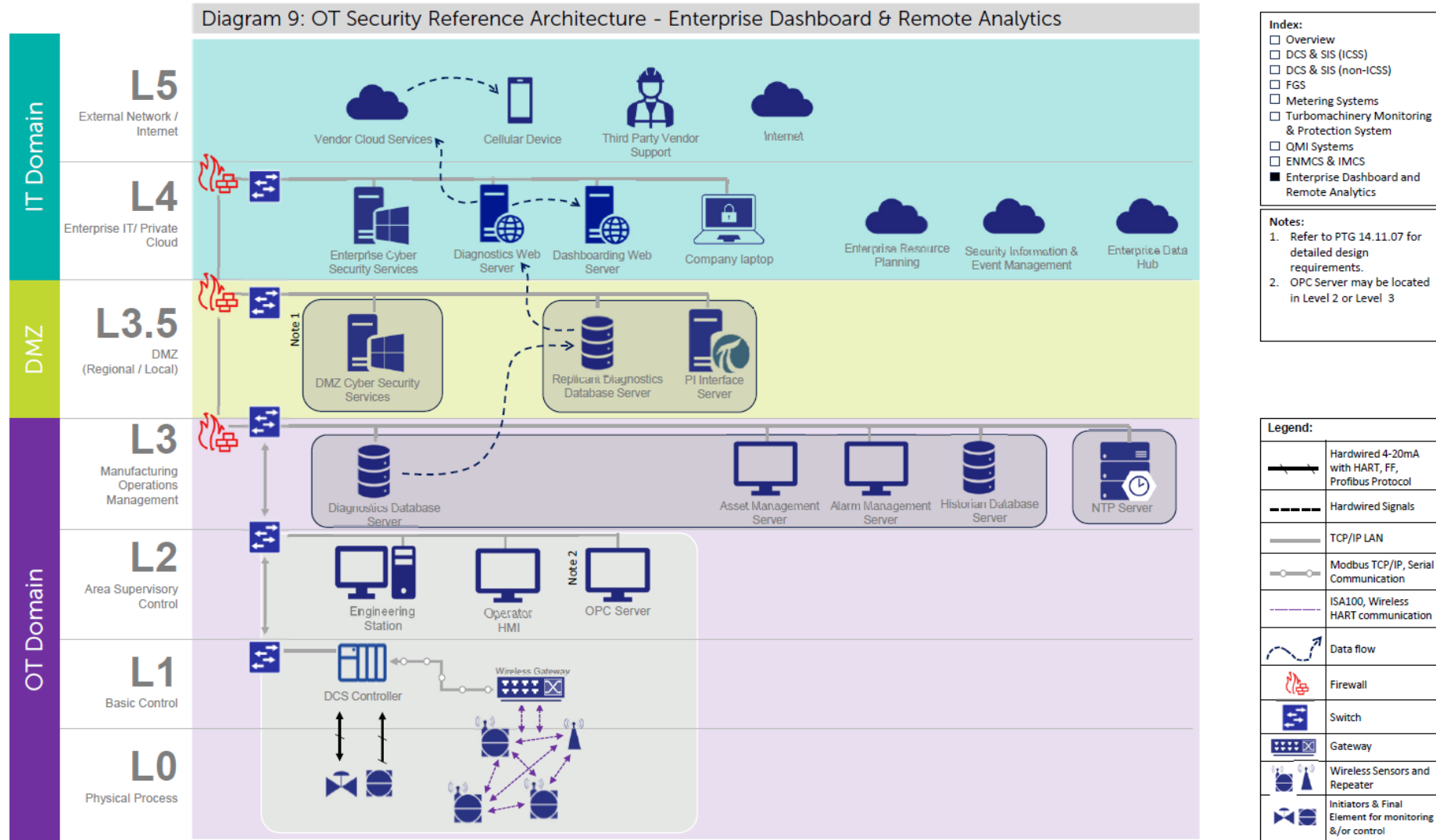
Digital twins act as a bridge between data analytics and AI, turning insights into actionable strategies for smarter decision-making.



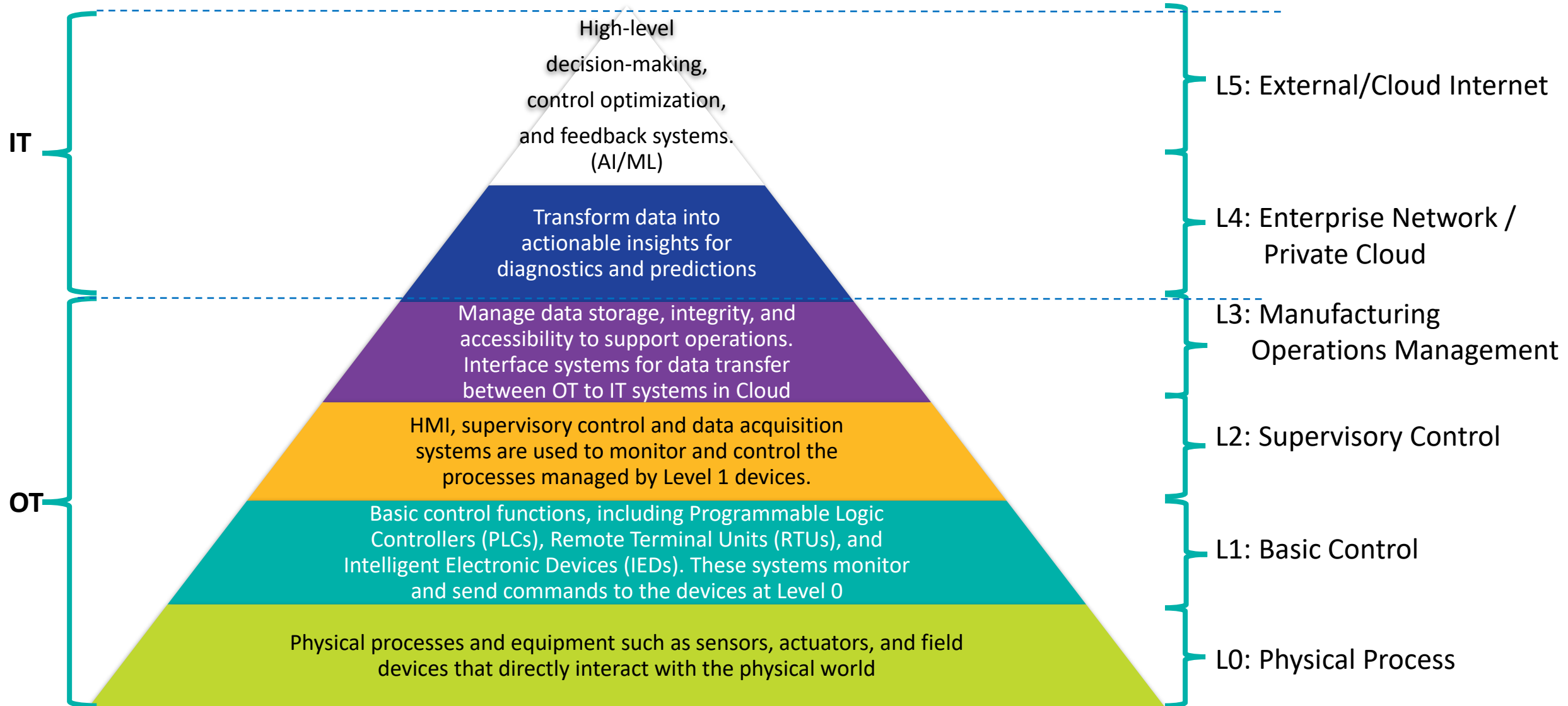
10.0 Data Communication in Digital Twin



10.1 Layers of Data Transfer from L0 to L5



10.1 Layers of Data Transfer from L0 to L5



10.2 Sensor Types and Technologies



Instrumented Process

Pressure Sensor

- Piezoelectric Type
- Capacitive Type
- Strain Gauge Type

Flow Rate Sensor

- Positive Displacement Sensor
- Mass Flow Sensor
- Velocity Measuring Flow Sensor

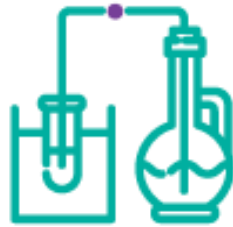
Temperature

- Thermistor
- RTD
- Thermocouple
- Infrared Thermal Sensor

Level Sensor

- Guided Wave Radar
- Displacer
- Capacitance
- Magnetic

QMI Analyzers



Position and Displacement Sensors

- Differential Transformer
- Potentiometer
- Strain Gauges

Proximity Sensors

- Inductive Proximity Sensor
- Capacitive Proximity Sensor

Velocity and Motion

- Tacho-Generator
- Encoder

Force

- Strain Gauge Load Cell
- Piezoelectric Sensor



Equipment conditioning

Noise & Vibration Sensor

- Piezoelectric-Based Accelerometer
- Capacitive Accelerometer
- Laser-Based Non-contact Vibrometer
- Acoustics

Current and Power Sensors

- Direct Sensing—Sensors Based on Ohm's Law
- Indirect Sensing—Sensors Based on Ohm's Law

Analyzers

- Moisture
- Gas Chromatograph
- Oxygen
- H₂S
- pH conductivity
- CEMS/PEMS

Gas Detectors

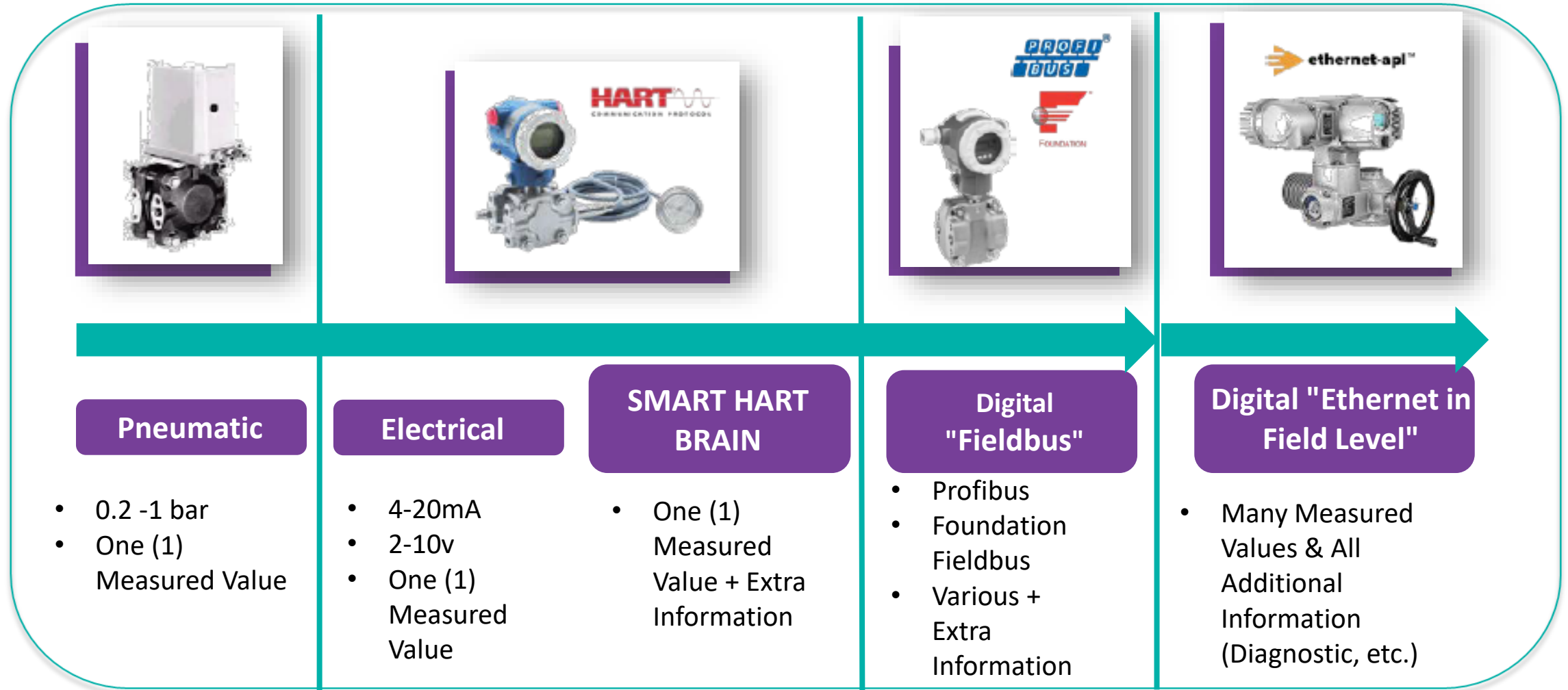
- Triple IR
- UV/IR
- CCTV Gas Cloud recognition

Environmental monitoring



10.2 Sensor Types and Technologies

Level 0 – Wireless Instrumentation



10.2 Sensor Types and Technologies

Level 0 – Wireless Instrumentation

Criteria		WirelessHART	ISA100	Cellular (5G/4G)	LoRa	Proprietary
Design	Original use case	Designed for the industrial automation space	Designed for the industrial automation space	Designed for cell and web communication with large scale implementation	Designed for automated meter reading (AMR), not instrumentation (single point distributed)	Typically sub 1 GHz
	Applications	Critical asset monitoring, Safety, Non-critical control	Critical asset monitoring, Safety, Non-critical control	Cell phones, Public networks, Remote monitoring	Low cost, Low rate, Wireless remote monitoring	Niche industry specific
Industrial Automation Features	Battery life	↑	→	→	↓	↓
	Reliability	↑	↓	↓	↓	↓
	Security	↑	↑	↑	↓	↓
	Scalability	↑	↑	→	→	↓
	Interoperability	↑	↓	↑	↓	↓
	Range	→	→	↑	↑	↑
	Update rate	↑	↑	→	↓	↓

10.3 Data Communication Technologies

Level 1/2 – ICSS and SCADA Typical Communication Protocols

MODBUS RTU/TCP

- A widely used protocol in industrial environments for serial communication between devices

OPC Unified Architecture (UA)

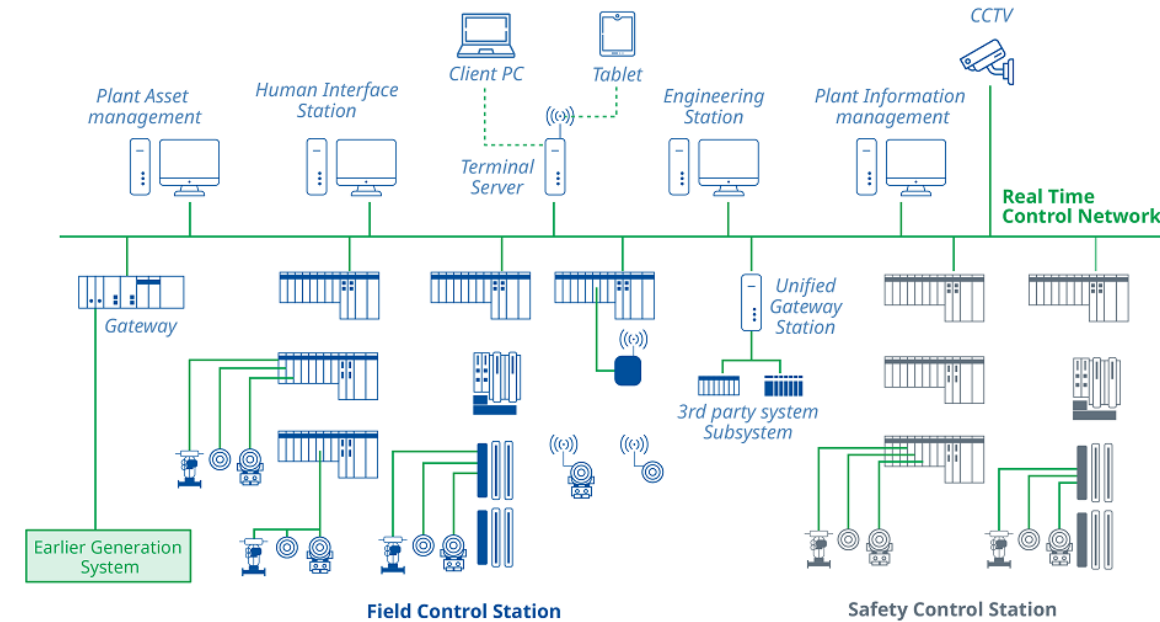
- Standardized, platform-independent protocol used for secure data exchange between industrial devices and systems, enabling seamless communication across different manufacturers and operating systems within industrial automation environments

Profinet/Ethernet-IP/HART-IP

- An open-source industry technical standard for data communication over Industrial Ethernet, designed for collecting data from, and controlling equipment in industrial systems, with a particular strength in delivering data under tight time constraints.

Distributed Network Protocol (DNP3)

- A protocol used primarily in utilities such as water and electric systems for real-time communication between SCADA systems and field devices

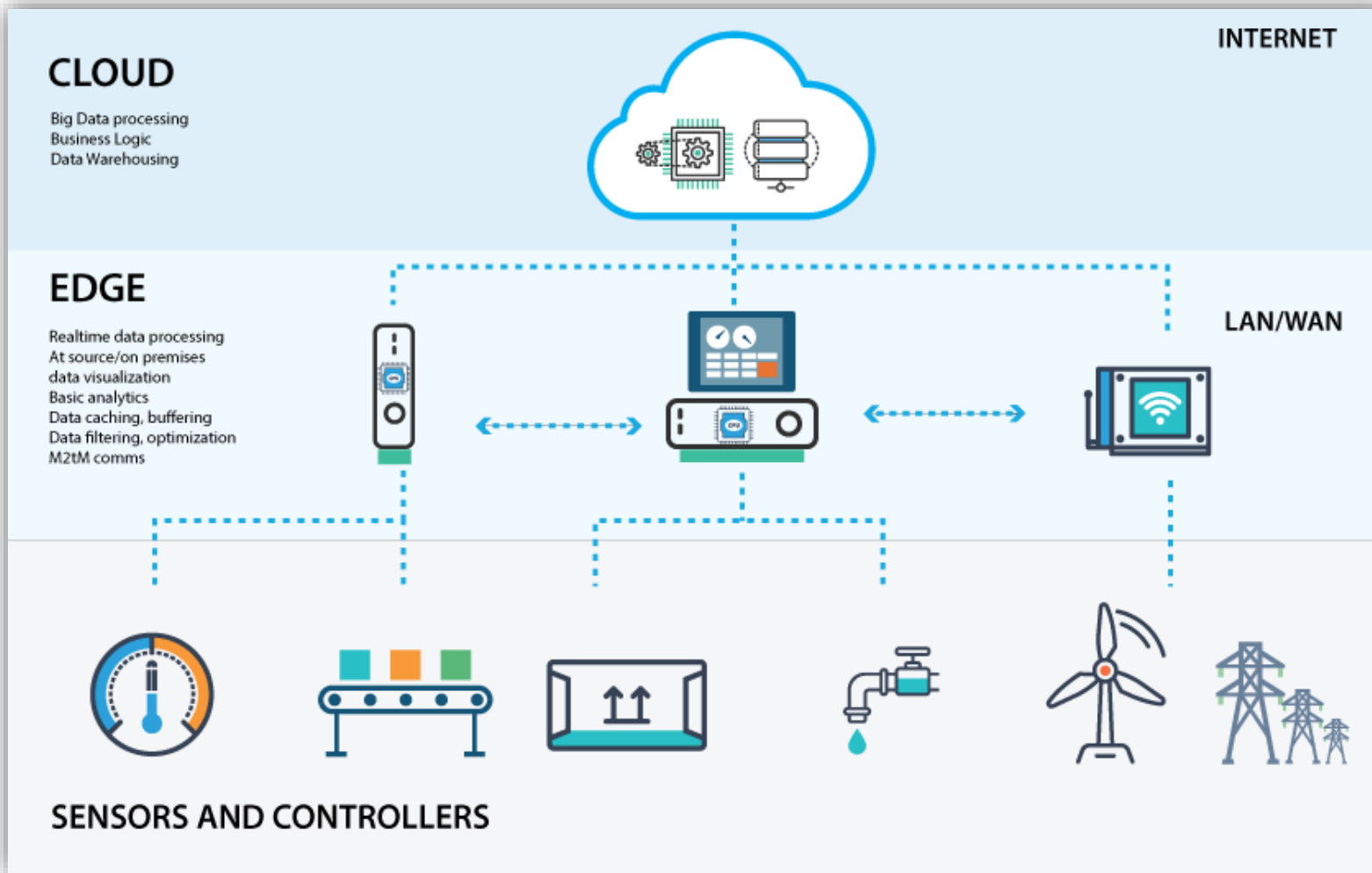


Proprietary DCS Protocols

- Developed and maintained by specific companies for use within their own DCS products. Typically, it is not openly licensed for use by other parties.
- E.g. Honeywell FTE, Emerson DeltaV, ABB HIE, Yokogawa Vnet/IP

10.3 Data Communication Technologies

Level 3/4 – Data transformation from OT to IT centric systems



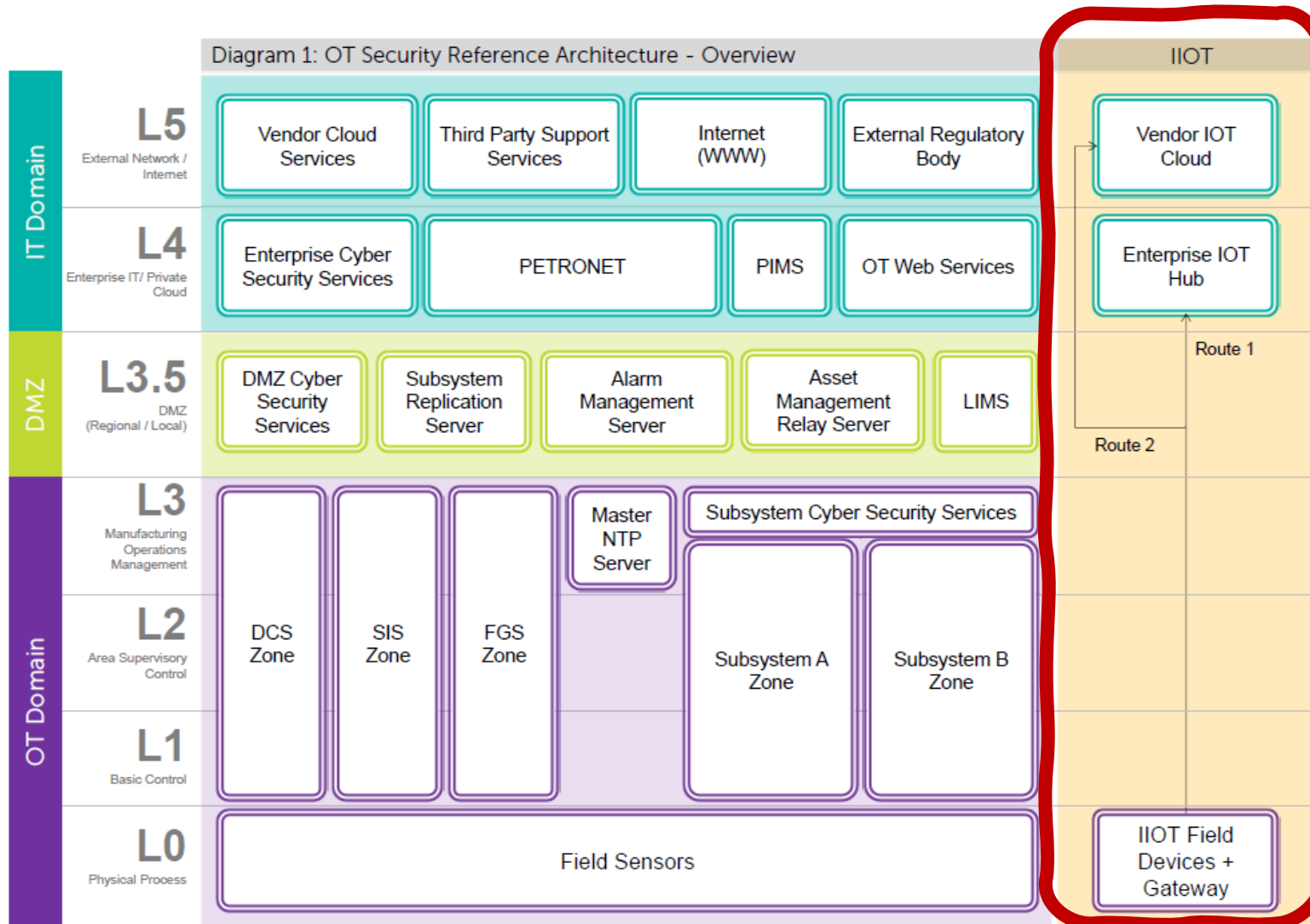
Plant Information Management Systems (PIMS)

- **Data Collection:** It collects high-frequency data from various sources, including sensors, control systems, and IoT devices.
- **Data Storage:** The system stores this data with high fidelity, ensuring no loss of critical information.
- **Data Analysis:** It provides tools for analyzing and visualizing data, helping users gain insights and make informed decisions.
- **Integration:** The PI System integrates with other IT and OT systems, enabling seamless data flow across the enterprise.

10.3 Data Communication Technologies

Level 3/4 – IIoT as a data communication alternative

Diagram 1: OT Security Reference Architecture - Overview



Industrial Internet of Things (IIoT)

- IIoT is a wireless technology that include several key protocols and standards that facilitate efficient and reliable data exchange:
 - **Message Queue Telemetry Transport (MQTT)**
 - **Data Distribution Service (DDS)**
 - **Advanced Message Queue Protocol (AMQP)**
 - **OPC Unified Architecture (OPC UA)**

10.4 Edge, Fog, Cloud Computing and Analytics



	Edge Computing	Fog Computing	Cloud Computing
Location of Data Processing	Data is processed directly at the source, such as on IoT devices or local edge servers	Data is processed in a distributed manner between the edge and the cloud, often on local network nodes or gateways	Data is processed and stored in centralized data centers, often far from the data source
Purpose	Reduces latency and bandwidth usage by processing data close to where it is generated	Extends the capabilities of edge computing by providing additional processing power and storage closer to the data source, but not directly on the devices	Provides scalable and flexible resources for data storage, processing, and management over the internet
Use Cases	Real-time applications like autonomous vehicles, industrial automation, and smart grids	Smart cities, connected healthcare, and large-scale IoT deployments where intermediate processing is beneficial	Data-intensive applications like big data analytics, web services, and enterprise IT infrastructure

10.4 Edge, Fog, Cloud Computing and Analytics

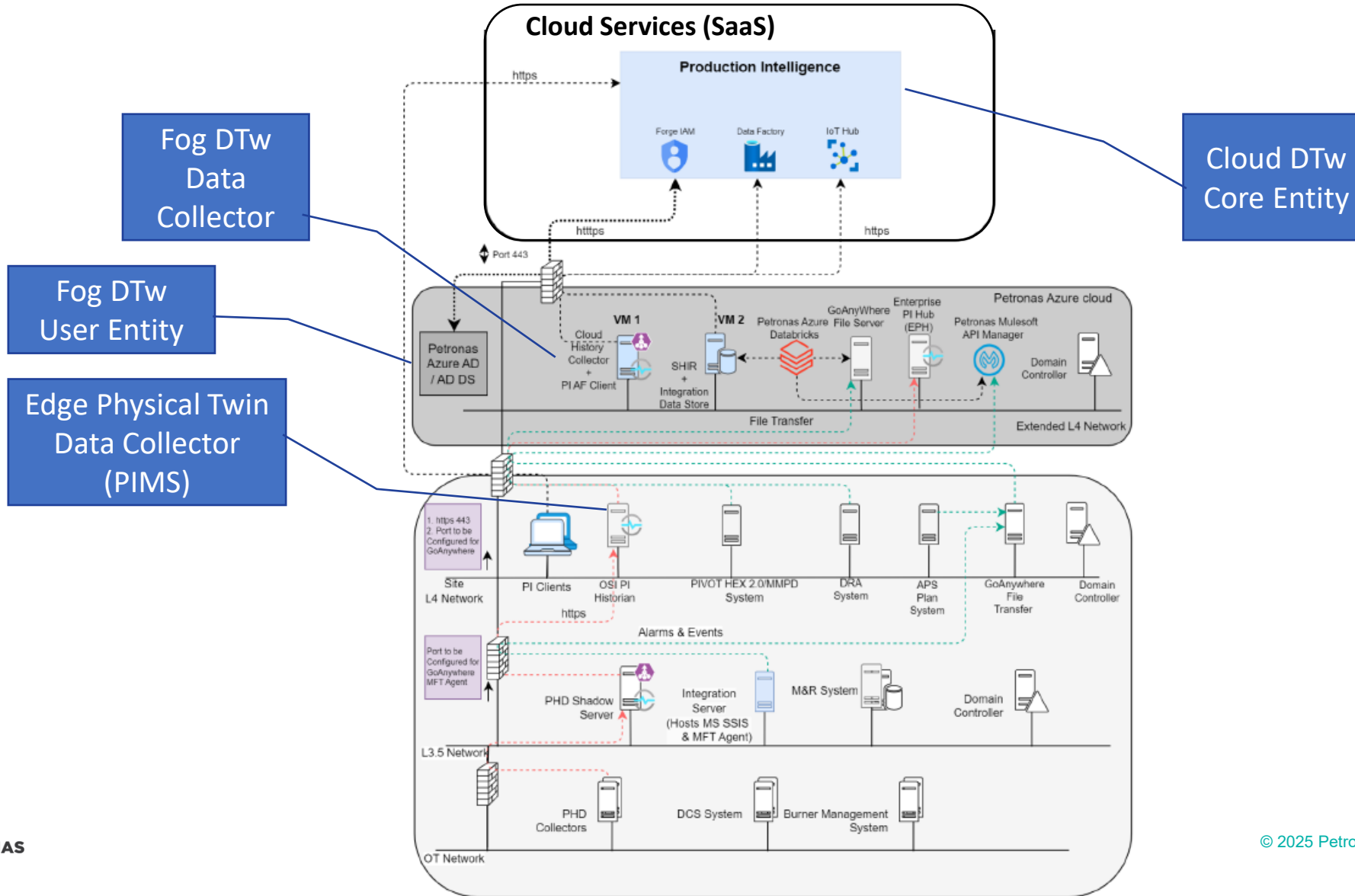
Comparison of Cloud, Fog and Edge Computing for Digital Twin

Function	Edge Computing	Fog Computing	Cloud Computing
Real-Time Processing	Ultra-low latency for immediate actions 	Near-real-time by aggregating local data 	Supports non-real-time analysis 
Scalability and Storage	Limited to device capacity 	Moderate scalability with local aggregation 	Highly scalable with vast storage 
Integration	Interfaces directly with physical systems 	Acts as a bridge between edge and cloud systems 	Centralized platform for global system integration 
Privacy and Security	Enhances privacy by keeping sensitive data at the source 	Improves data security by local processing whilst reducing complexity 	Requires complex cybersecurity features and higher security risks 
Cost	Higher capital expenditures to acquire the required hardware and software 	Balances costs by distributing load between edge and cloud 	Cost-effective due to pay-as-you-use pricing 

10.4 Edge, Fog, Cloud Computing and Analytics

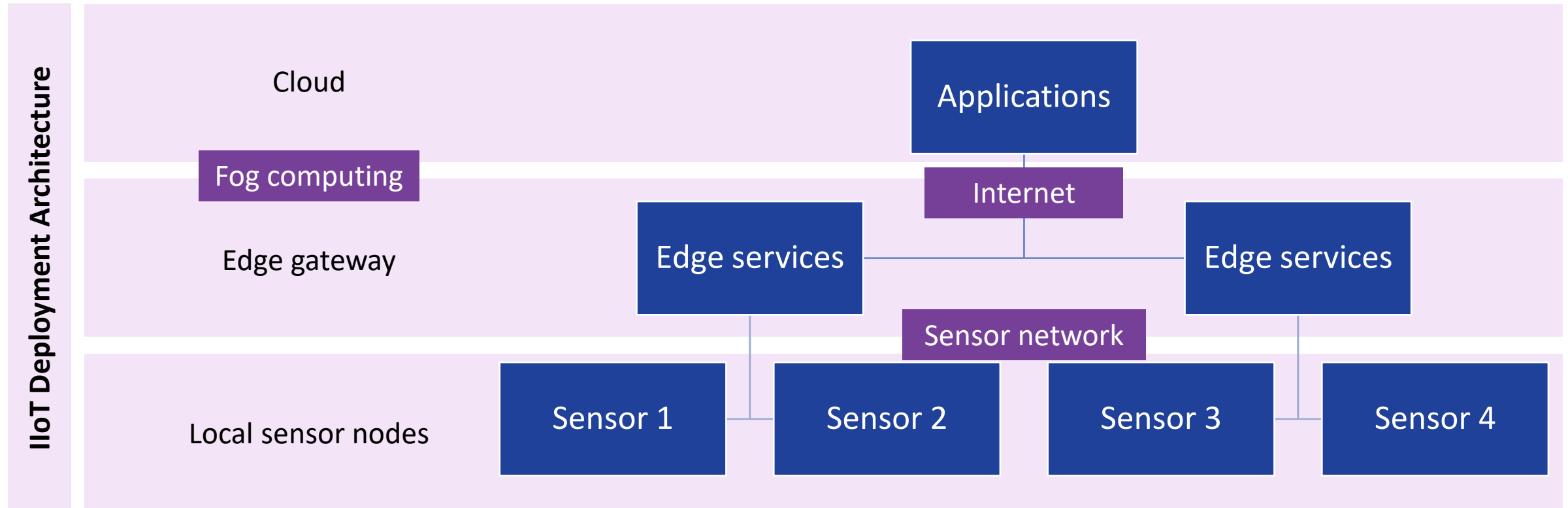
Network Architectures for Edge/Cloud Computing

ICSS Deployment Architecture



10.4 Edge, Fog, Cloud Computing and Analytics

Network Architectures for Edge/Cloud Computing



11.0 Digital Twin Applications



11.1

Case for Change Requiring
Digital Twin Application



11.2

Use Cases of Digital Twin in O&G



11.3

How Digital Twin Contributes
to Sustainability

11.1 Case for change

Cost Optimization

- Reduce operational and maintenance costs by simulating real-time scenarios.
- Enhance resource allocation and energy efficiency across facilities.
- Minimize unplanned downtime with predictive analytics.

Reliability Improvement

- Monitor asset health and predict failures before they occur.
- Optimize equipment performance through real-time insights.
- Enable faster decision-making with accurate, data-driven models.

Sustainability Goals

- Drive carbon reduction initiatives by identifying inefficiencies.
- Optimized production processes to lower environmental impact.

Innovation & Competitiveness

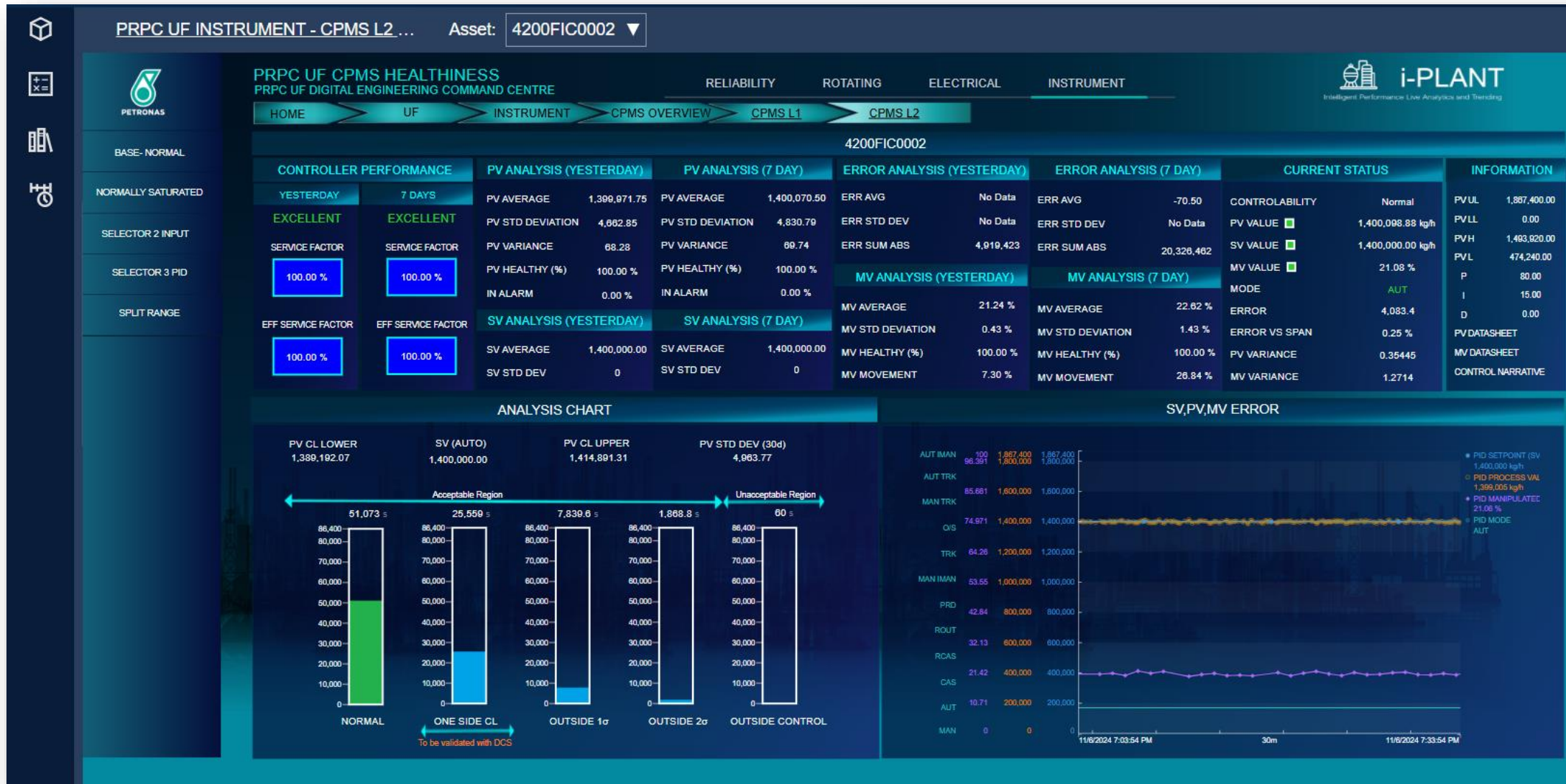
- Accelerate R&D and project implementation through virtual testing.
- Benchmark global best practices and deploy agile strategies.

Enhanced Collaboration

- Integrate data across teams for unified operational views.
- Enable remote monitoring and diagnostics for field and offshore operations.

11.2 Use Case 1 - I&C Implementation

PRPC UF Controller Performance Monitoring System



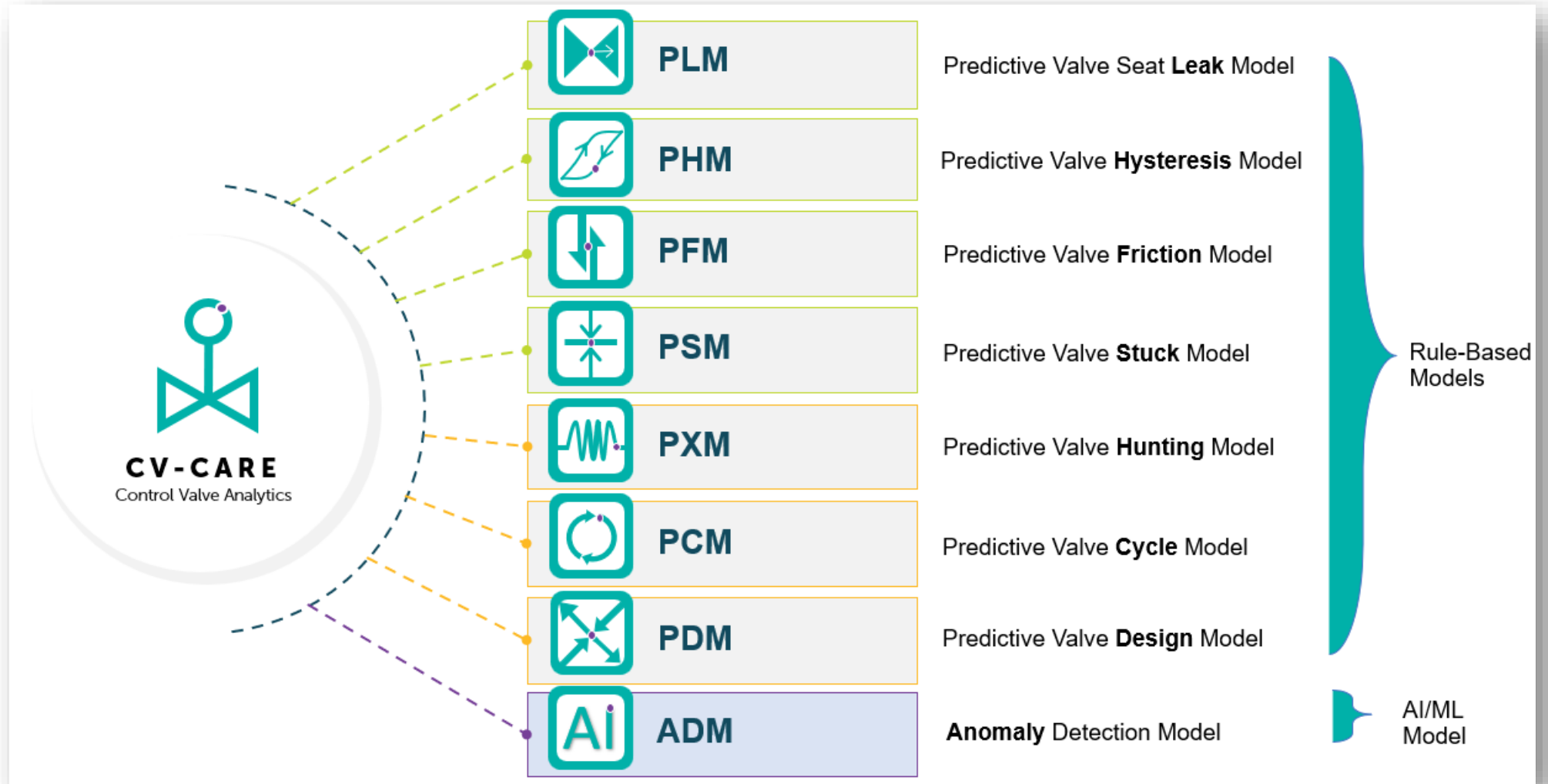
11.2 Use Case 2 - I&C Implementation

PRPC UF Turbomachinery Healthiness Monitoring



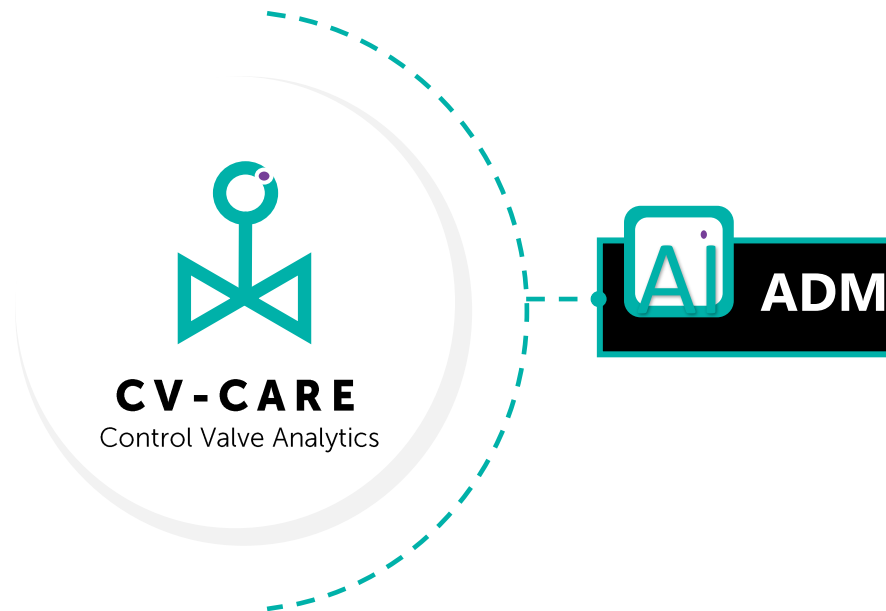
11.2 Use Case 3 - I&C Implementation

IRIS CV-CARE

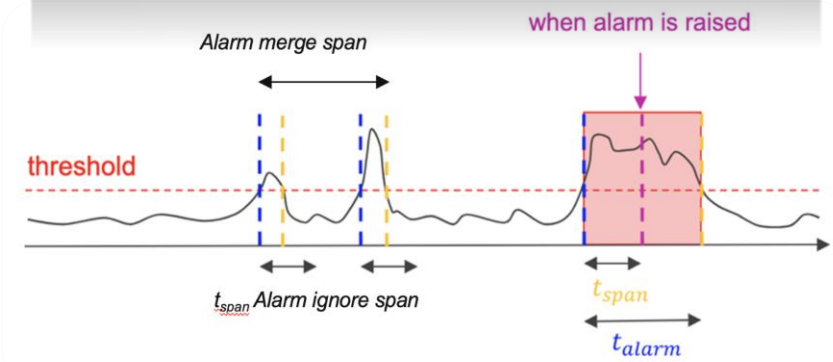
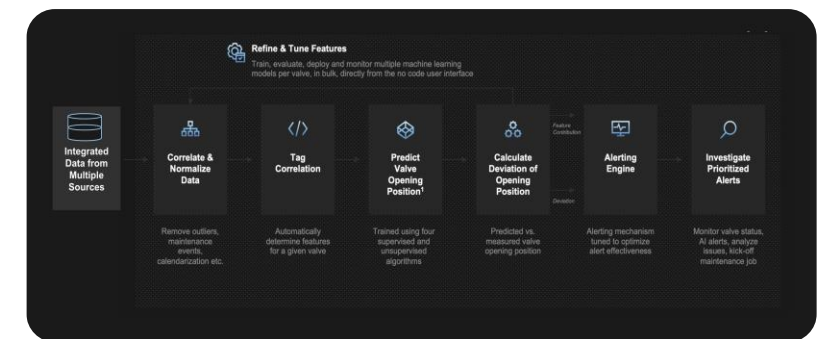


11.2 Use Case 4 - I&C Implementation

IRIS CV-CARE



Predict CV position using semi supervised approach with regression modeling to generate alert based on comparison against CV position based on predetermined criteria.



11.2 Use Case 5 - I&C Implementation

PGB GPU Machinery Monitoring Prediction Detection

PGB Dashboard

Overview

Equipment ▾

System 1

Last updated on 13/2/2025 10:05:00

Generate Report

OVERALL STATUS

GPS

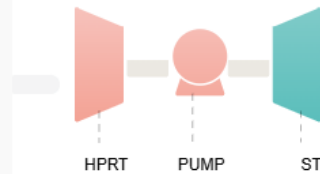
EQUIPMENT

INSTRUMENT



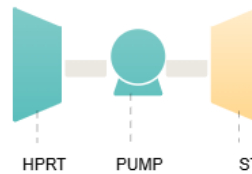
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RUNNING



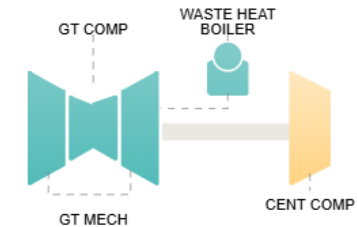
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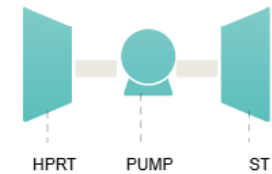
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RUNNING



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RUNNING



GPK

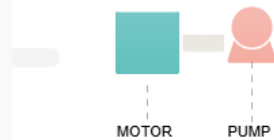
EQUIPMENT

INSTRUMENT



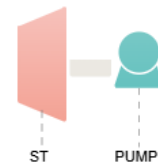
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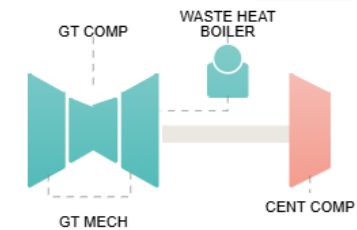
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RUNNING



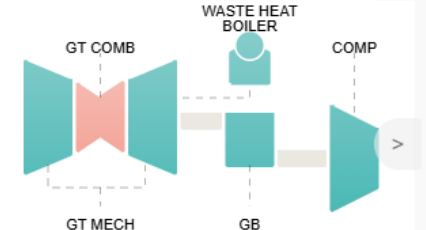
RT4-501

RUNNING



RT2-701

RUNNING



11.2 Use Case 6 -I&C Implementation

PGB GPU Machinery Monitoring Prediction Detection

AVEVA™ Predictive Analytics

Trends Faults Forecast



02/03/2025 10:18 to 02/13/2025 10:18 at 5 Minutes

Time Span ☐ Auto (Last: 10:18)

10



Days



✓ PGB GPU GTF GB MECH

✓ (Overall Model Residual)  

✓ LSGB SHAFT SPEED (PGB...



✓ HSGB SHAFT SPEED (PG...



✓ GB BRG CASING VIB (PGB...



✓ GB BRG CASING VIB 2 (P...



✓ LSGB BRG DE VIB X (PGB...



✓ LSGB BRG DE VIB Y (PGB...



✓ LSGB BRG DE VIB X 2 (PG...



✓ LSGB BRG DE VIB Y 2 (PG...



✓ LSGB BRG NDE VIB X (PG...



✓ LSGB BRG NDE VIB Y (PG...



Filters

Threshold

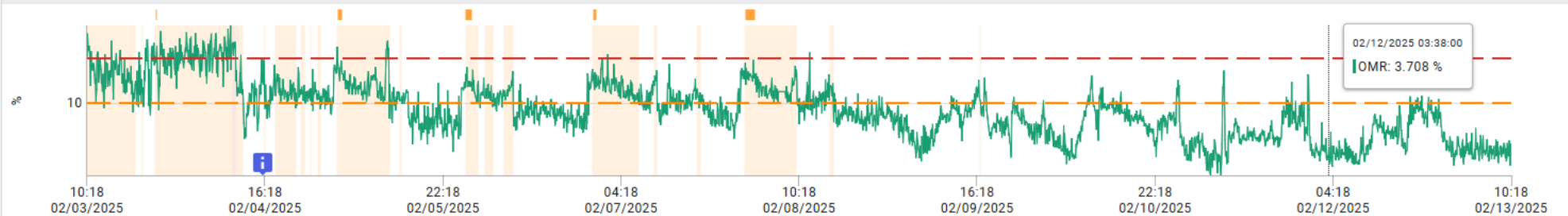
Alerts

Threshold, Mark

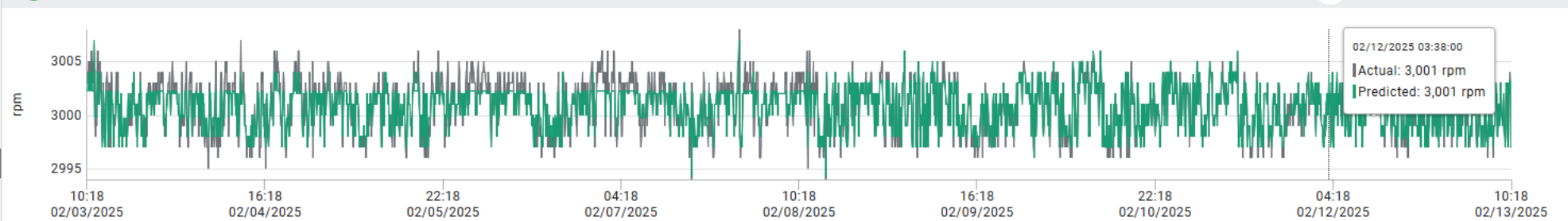
Quality

Mark, Hide

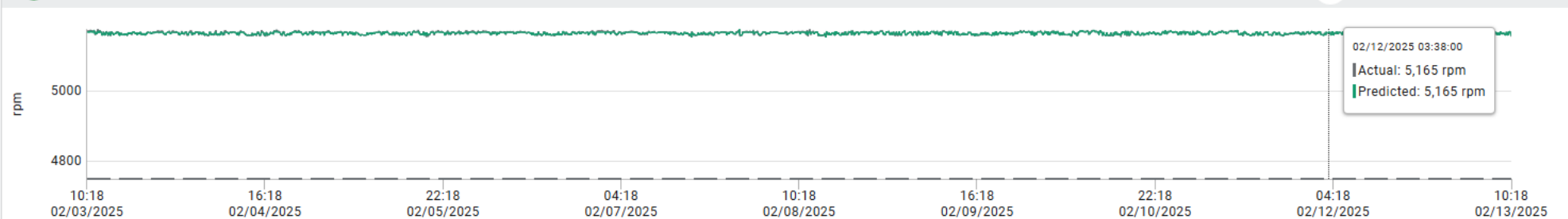
  (Overall Model Residual)



  LSGB SHAFT SPEED (PGB.GPU.UK.GT-F.MVUK_KF.77RP-9-1)

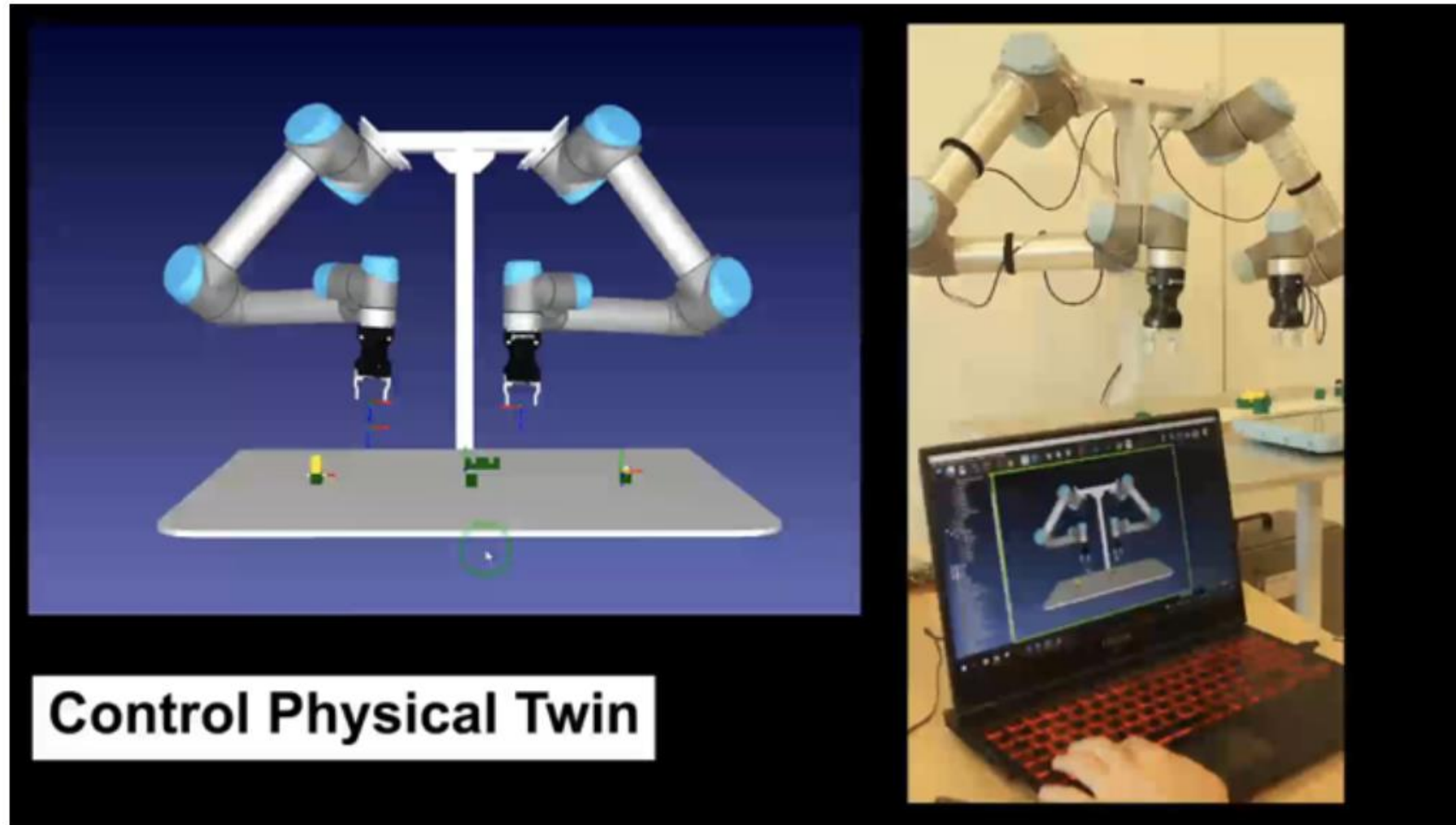


  HSGB SHAFT SPEED (PGB.GPU.UK.GT-F.MVUK_KF.77RP-1-1)

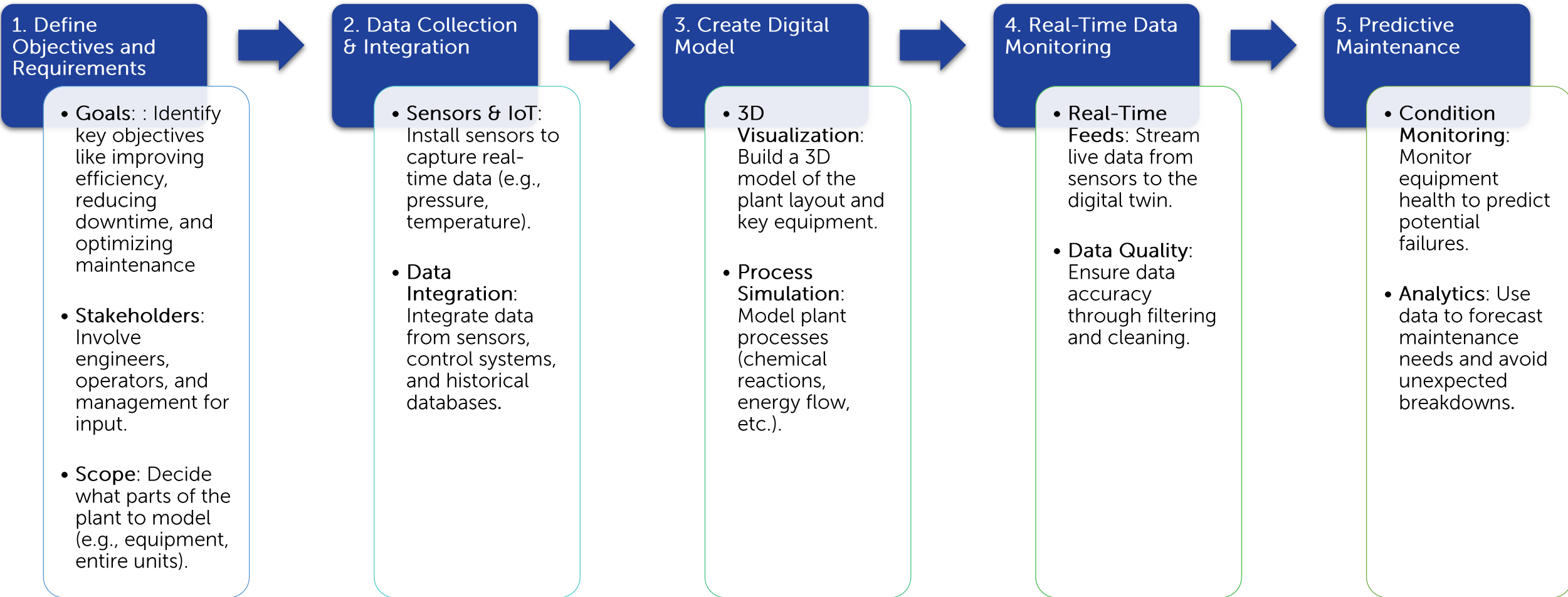


11.2 Use Case 7 3D Model

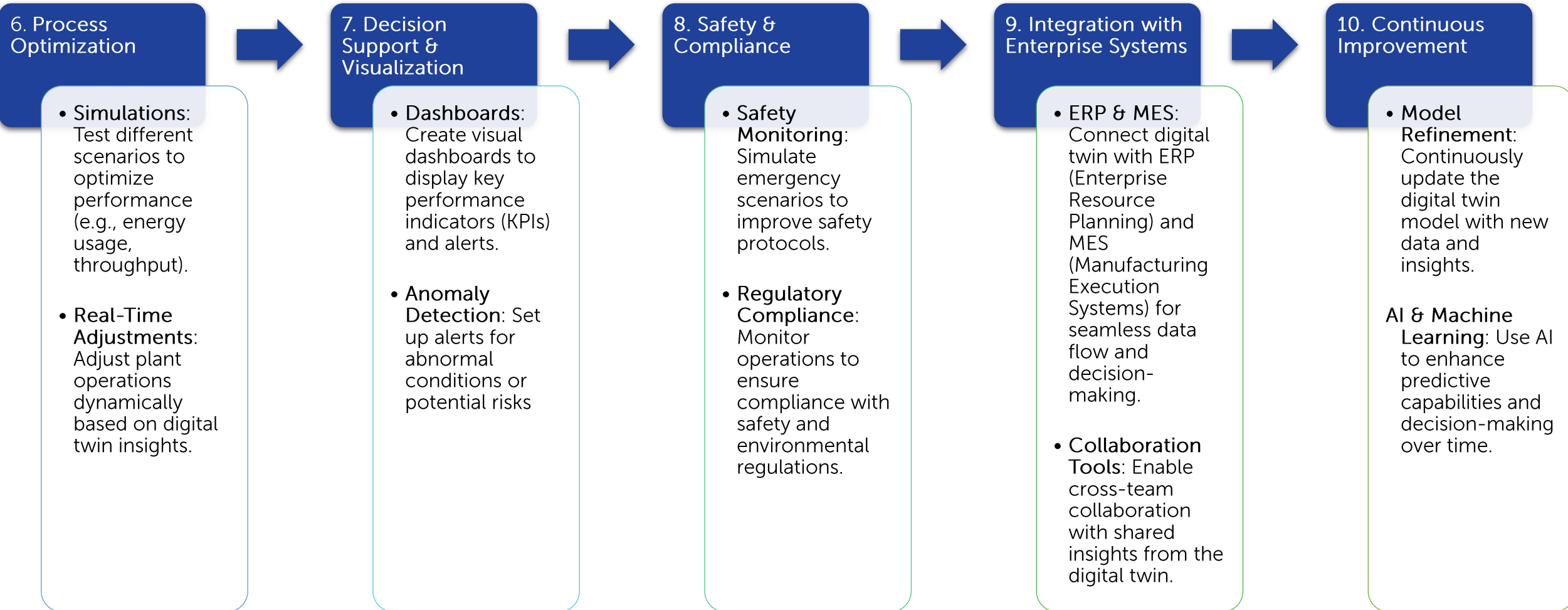
Robot Manipulator in 3D Drawing for the control of Physical Twin



Solution Approach For Digital Twin In A Process Plant



Solution Approach For Digital Twin In A Process Plant



Digital Twin and Sustainability



Energy Optimization

- Reduce energy waste with real-time system adjustments.
- Digital twins **monitor and simulate systems** in real time to reduce energy consumption and maximize efficiency.



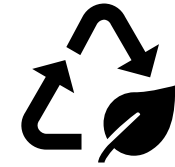
Predictive Maintenance

- Lower resource use and waste by extending equipment life.
- Digital twins **predict equipment failures, enabling timely interventions** and extending asset lifecycles.



Carbon Footprint Reduction

- Simulate and mitigate environmental impacts of operations.
- Digital twins **track and optimize processes** to minimize carbon emissions and improve environmental compliance.



Circular Economy Support

- Enhance recycling and reuse strategies.
- Digital twins **analyze product lifecycles** to enhance recyclability, resource recovery, and sustainable design.

